Adding (1) and (2), we get

$$2I = \int_2^3 \frac{\sqrt{x}+\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}}\,dx = \int_2^3 dx = [x]_2^3 = 3-2 = 1.$$

Hence, we get $I = \frac{1}{2}$.

Example 9.30

Evaluate $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x}\,dx$

Solution

$$\text{Let } I = \int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x}\,dx \qquad \ldots (1)$$

Using $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$ we get,

$$I = \int_{-\pi}^{\pi} \frac{\cos^2(\pi-\pi-x)}{1+a^{\pi-\pi-x}}\,dx$$

$$= \int_{-\pi}^{\pi} \frac{\cos^2(-x)}{1+a^{-x}}\,dx$$

$$= \int_{-\pi}^{\pi} a^x \left(\frac{\cos^2 x}{a^x+1}\right) dx \qquad \ldots (2)$$

Adding (1) and (2) we get

$$2I = \int_{-\pi}^{\pi} \frac{\cos^2 x}{a^x+1}\,(a^x+1)\,dx = \int_{-\pi}^{\pi} \cos^2 x\,dx$$

$$= 2\int_0^{\pi} \cos^2 x\,dx \text{ (since } \cos^2 x \text{ is an even function)}$$

$$\text{Hence } I = \int_0^{\pi} \frac{(1+\cos 2x)}{2}\,dx = \frac{1}{2}\left[x+\frac{\sin 2x}{2}\right]_0^{\pi} = \frac{1}{2}[\pi] = \frac{\pi}{2}.$$

EXERCISE 9.3

1. Evaluate the following definite integrals :

(i) $\int_3^4 \frac{dx}{x^2-4}$ (ii) $\int_{-1}^1 \frac{dx}{x^2+2x+5}$ (iii) $\int_0^1 \sqrt{\frac{1-x}{1+x}}\,dx$

(iv) $\int_0^{\frac{\pi}{2}} e^x \left(\frac{1+\sin x}{1+\cos x}\right) dx$ (v) $\int_0^{\frac{\pi}{2}} \sqrt{\cos\theta}\,\sin^3\theta\,d\theta$ (vi) $\int_0^1 \frac{1-x^2}{(1+x^2)^2}\,dx$

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